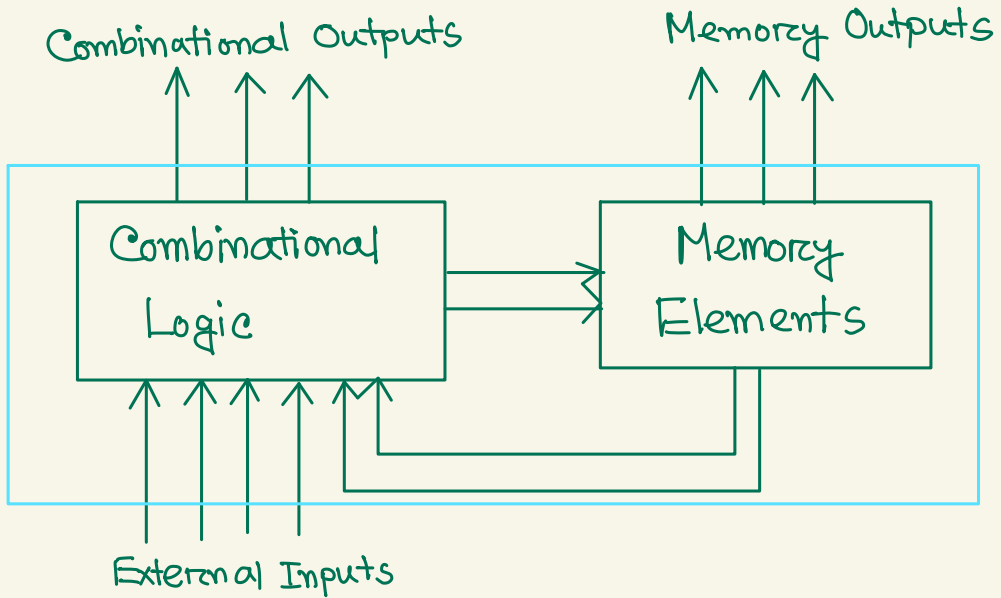




Sequential Circuit

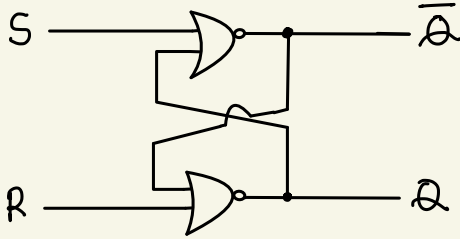
Flip-Flop



- Sequential Circuit = Combinational Logic + Memory Elements
- Output = External Input + Present state of memory element

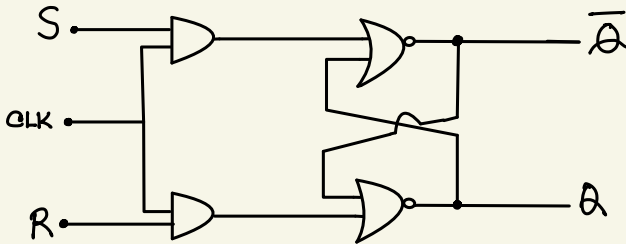
SR Flipflop:

SR Latch: (NOR gate)



S	R	Q(t+1)
0	0	No change
0	1	0 (Reset)
1	0	1 (Set)
1	1	Invalid

SR Flipflop:

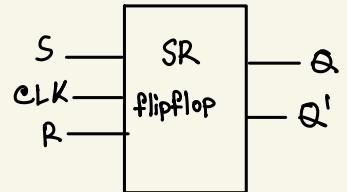


* Truth Table:

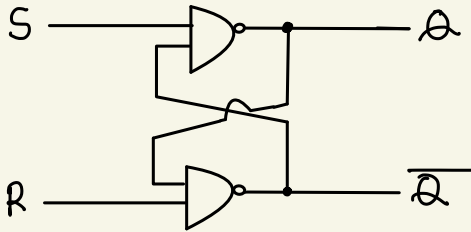
S	R	Q(t+1)	Q(t+1)'
0	0	No Change	
0	1	0	1
1	0	1	0
1	1	Invalid	

Reset

Set

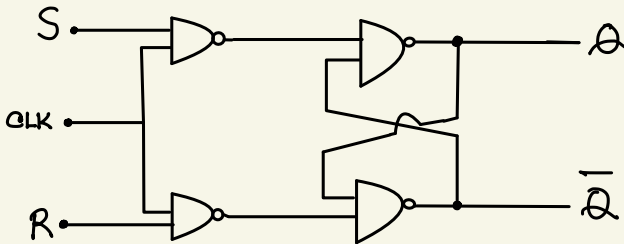


SR Latch: (NAND gate)



S	R	$Q(t+1)$
0	0	Invalid
0	1	1 (Set)
1	0	0 (Reset)
1	1	No change

SR Flipflop:

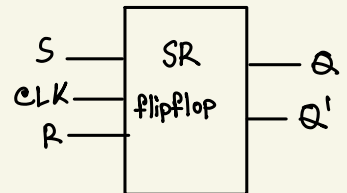


* Truth Table:

S	R	$Q(t+1)$	$Q(t+1)'$
0	0	No Change	
0	1	0	1
1	0	1	0
1	1	Invalid	

Reset

Set



* Characteristics Table:

S	R	Operation	$Q(t)$	$Q(t+1)$
0	0	No change	0	0
			1	1
0	1	Reset	X	0
1	0	Set	X	1
1	1	Invalid	—	—

optional way

$Q(t)$	S	R	$Q(t+1)$
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	indeterminate X
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	indeterminate X

• Characteristics Equation:

$$Q(t+1) = S + R'Q$$

• $S \cdot R = 0$

K-map:

	$S'R'$	$S'R$	SR	SR'
$Q(t)'$	0	1	X ₃	1 ₂
$Q(t)$	1 ₄	5	X ₇	1 ₆

*Excitation Table:

$Q(t)$	$Q(t+1)$	Operations		Combined operation	
		S	R	S	R
0	0	0	0	0	X
		0	1		
0	1	1	0	1	0
1	0	0	1	0	1
1	1	0	0	X	0
		1	0		

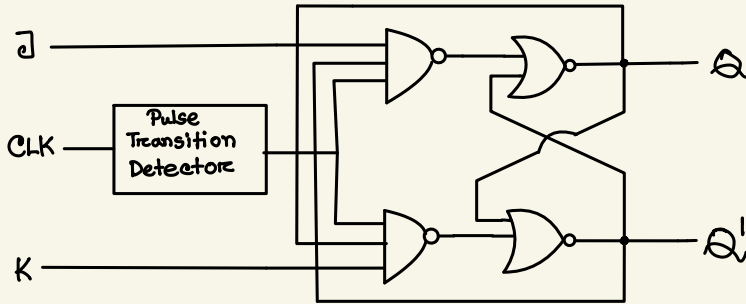
optional way

$Q(t)$	$Q(t+1)$	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

- No change $\rightarrow Q(t) = Q(t+1)$
- Set $\rightarrow Q(t+1) = 1$
- Reset $\rightarrow Q(t+1) = 0$

JK Flipflop:

JK Flipflop Diagram:

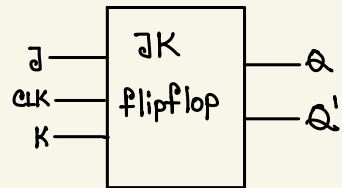


*Truth Table:

J	K	$Q(t+1)$	$Q(t+1)'$
0	0	No Change	
0	1	0	1
1	0	1	0
1	1	Toggle	

Reset

Set



*Characteristics Table:

J	K	Operation	Q(t)	Q(t+1)
0	0	No change	0	0
			1	1
0	1	Reset	X	0
1	0	Set	X	1
1	1	Toggle	0	1
			1	0

Optional

Q(t)	J	K	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	0

• Characteristics Equation:

$$Q(t+1) = J \cdot Q' + K' \cdot Q$$

K-map:

	J'K'	J'K	JK	JK'
Q(t)'	0	1	1	1
Q(t)	1	5	7	6

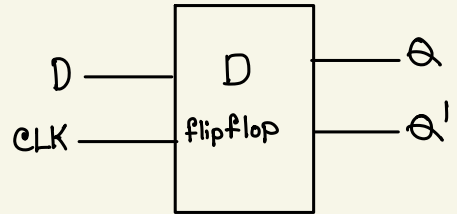
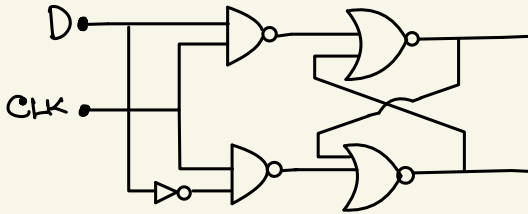
* Excitation Table:

$Q(t)$	$Q(t+1)$	Operations		Combined Operations	
		$\overline{Q}$	K	$\overline{Q}$	K
0	0	0	0	0	X
		0	1		
0	1	1	0	1	X
		1	1		
1	0	0	1	X	1
		1	1		
1	1	0	0	X	0
		1	0		

- No change
 $\hookrightarrow Q(t) = Q(t+1)$
- Set
 $\hookrightarrow Q(t+1) = 1$
- Reset
 $\hookrightarrow Q(t) = 0$
- Toggle
 $\hookrightarrow Q(t+1) = Q(t)'$

D Flipflop:

D Flipflop Diagram:



* Truth Table:

D	D'	Q(t+1)	Q(t+1)'
0	1	0	1
1	0	1	0

* Characteristic Table:

D	Operation	$Q(t)$	$Q(t+1)$
0	X	0	0
		1	0
1	X	0	1
		1	1

Optional

$Q(t)$	D	$Q(t+1)$
0	0	0
0	1	1
1	0	0
1	1	1

• Characteristics Equation:

$$Q(t+1) = D$$

k-map:

	D'	D
$Q(t)'$	0	1 ₁
$Q(t)$	2	1 ₃

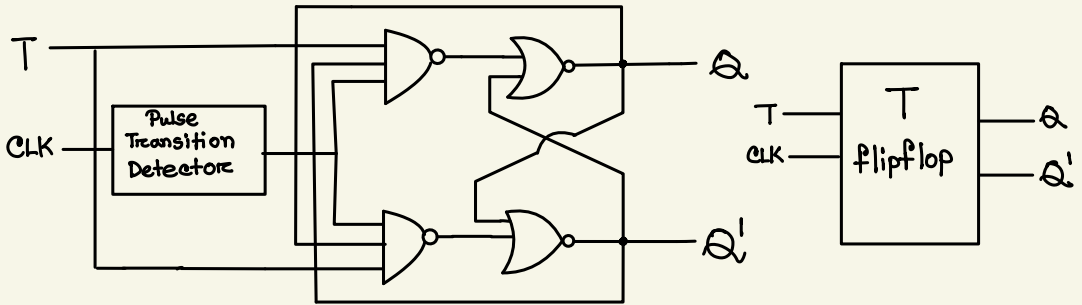
* Excitation Table:

$Q(t)$	$Q(t+1)$	D
0	0	0
0	1	1
1	0	0
1	1	1

• $Q(t+1) = D$

T-Flipflop:

T-Flipflop Diagram:



*Truth Table:

T	$Q(t+1)$	$Q(t+1)'$
0	No change	
1	Toggle	

* Characteristics Table:

T	Operation	$Q(t)$	$Q(t+1)$
0	No change	0	0
		1	1
1	Toggle	0	1
		1	0

Optional

$Q(t)$	T	$Q(t+1)$
0	0	0
0	1	1
1	0	1
1	1	0

• Characteristics Equation:

∴ Equation:

$$Q(t+1) = TQ' + T'Q$$

K-map:

	T'	T
$Q(t)'$	0	$\boxed{1}_1$
$Q(t)$	$\boxed{1}_2$	3

* Excitation Table:

$Q(t)$	$Q(t+1)$	T
0	0	0
0	1	1
1	0	1
1	1	0

- No change $\rightarrow Q(t) = Q(t+1)$
- Toggle $\rightarrow Q(t) \neq Q(t+1)$

Excitation Tables

Q	Q^+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

JK Flip-flop

Q	Q^+	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

SR Flip-flop

Q	Q^+	D
0	0	0
0	1	1
1	0	0
1	1	1

D Flip-flop

Q	Q^+	T
0	0	0
0	1	1
1	0	1
1	1	0

T Flip-flop